

Full support signed permutations and their asymptotic coefficients

2 October 2026

Abstract

We prove the explicit large-order conjecture recorded in OEIS A260952 and derive every fixed correction order. The coefficients belong to the factorial asymptotics of full-support elements of Weyl groups of types B and D . An exact positive Stirling transform identifies the dominant scale $\log 3$ and gives a proof with controlled remainders. For the original signed-permutation model, two exact endpoint sums admit separate expansions uniformly near the zero-sign-weight boundary. They yield an all-order sparse-sign crossover and a Poisson law conditioned to be positive. Growth-index inverses are provided with an explicit rounding qualification. The underlying generating functions, connected-permutation asymptotics, and general Stirling-transform machinery are prior work; the contribution is the specialized conjecture resolution, its refinements, and the uniform crossover.

1 The conjecture and the normalization

Let

$$g(x) = \sum_{n \geq 0} n! x^n, \quad C(x) = 1 - \frac{1}{g(x)} = \sum_{n \geq 1} c_n x^n, \quad B_n(q) = [x^n] \frac{g(qx)}{g(x)}. \quad (1)$$

These are formal power series. The integers c_n count indecomposable permutations and form OEIS [A003319](#). Put $b_n = B_n(2)$, which is [A109253](#), the number of elements of the Weyl group B_n whose support contains every standard simple reflection. For type D , the corresponding sequence [A112225](#), denoted by d_n , satisfies the classical identity

$$d_n = \frac{b_n - 3c_n}{2}, \quad n \geq 2. \quad (2)$$

Bergeron, Hohlweg, and Zabrocki [\[1\]](#) give the generating functions and their combinatorial interpretation. Martin and Kearney [\[2\]](#) give integral representations.

For fixed $q > 1$, define $p_0(q) = 1$ and the asymptotic coefficients $p_m(q)$ by

$$\frac{B_n(q)}{q^n n!} \sim \sum_{m \geq 0} \frac{p_m(q)}{n^m}. \quad (3)$$

The sequence $a_m = \text{A260952}(m)$ uses powers of $1/(2n)$ instead. Consequently

$$a_m = 2^m p_m(2). \quad (4)$$

Kotěšovec's 2015 conjecture, as stated in the OEIS entry, is

$$a_m \sim -\frac{2^{m+1} m!}{9(\log 3)^{m+1}}. \quad (5)$$

The factor 2^m in [\(4\)](#) must be retained when comparing the two normalizations.

Theorem 1. *The conjecture (5) holds. More precisely, with $\tau = \log 3$ and $(m)_r = m(m-1)\cdots(m-r+1)$,*

$$a_m = -\frac{2^{m+1}m!}{9\tau^{m+1}} \left\{ 1 - \frac{2\tau}{m} - \frac{9\tau^2}{4(m)_2} - \frac{27\tau^3}{2(m)_3} - \frac{1755\tau^4}{16(m)_4} + O(m^{-5}) \right\}. \quad (6)$$

Every fixed subsequent order is computable by a finite formula with the corresponding next-order remainder. Moreover, $a_m < 0$ for every $m \geq 1$.

The next two coefficients in this falling-factorial form are

$$-\frac{9099\tau^5}{8(m)_5}, \quad -\frac{907821\tau^6}{64(m)_6}.$$

They are obtained symbolically, rather than fitted to numerical data. The expansion is not asserted to converge.

[write] Added 2 October 2026, batch 77. This report is a single manuscript, printed in full: manuscript 02 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `Coxeter_Full_Support_Asymptotics.zip` (wrapper directory `Coxeter_Full_Support/`, main file `Coxeter_Full_Support_Asymptotics.tex`, now `article.tex`) in commit 096ee7b87, and placed in commit d0e6008d9 in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript names no ProveIt commit and describes no search of the repository, so it has no pin; its author line is empty. The bounded source search of Section 9 is the manuscript’s own literature search; the intake’s untruncated search of the repository (the research-report collection, the transseries volumes and `0eis/`) found no other treatment of A260952, A109253 or A112225, or of full-support elements of Coxeter groups. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step prefixed every label with `fss:`, added section labels and these **[write]** notes, and three references [7, 8, 9], and set the bibliography ragged-right; no statement, proof or number of the manuscript was altered. The type- B/D models, generating functions and connectivity characterization remain credited to [1], and the shifted-Stirling and compound-Poisson method to [2], exactly as delivered.

A neighbouring report. The same batch delivered (as manuscript 17) an all-orders study of the distinct parabolic double cosets of the symmetric group (OEIS A260700), now the collection report `a260700-parabolic-double-cosets` [7], which proves Browning’s higher-order conjecture. Both reports count objects defined by the standard parabolic subgroups of a Coxeter group — here the elements of the Weyl groups B_n and D_n lying in no proper standard parabolic subgroup, there the distinct double cosets $W_I w W_J$ in type A — and both reach a factorial scale through a Stirling transform and invert it with a Lambert W seed. Their theorems, generating functions and scales differ ($\tau = \log 3$ and $m!/(\log 3)^{m+1}$ here; $\rho = \log 2$ and $n!\rho^{-2n}$ there), and neither report uses the other.

Notation. Each symbol is printed as delivered. The following letters carry two meanings here, or a different meaning in [7].

symbol	meaning here (and what it is <i>not</i>)
$B_n, B_n(q)$	the Weyl group of type B , and the sign-weighted full-support count (1), $b_n = B_n(2)$. Not the correction polynomials $B_m(r)$ of [7].
$p_m(q), a_m$	$p_m(q)$ are <i>asymptotic coefficients</i> (3), $a_m = 2^m p_m(2)$ is A260952 (4); $a = (1+q)/q$ in Section 6. In [7], p_n counts double cosets.
c_n	indecomposable permutations (A003319); not the constant c of Browning's conjecture in [7].
τ	$\tau = \log 3$ in Theorem 1; $\tau = \log(1+q)$ in Section 6 and in $\tau(v) = \log(1+q/v)$; the same at $q = 2$.
D	type D with d_n (A112225); $D(n) = c_n/n!$ (14) and its formal version $D(x)$; $D_0 = \log(\alpha n_0)$ in Section 8.
$L, H, \mathcal{H}$	the endpoint sums $L_n(q)$, $H_n(q)$ and their generators $L(q, x)$, $H(q, x)$; $\mathcal{H}(t)$ is the binary entropy.
K, K_n	the carrier constant K of (37); K_n the number of negative signs (a random variable). Neither is the K of [7].
A_1, A_2, A_0	pole coefficients in Section 6; $A_0 = \frac{1}{2} \log(2\pi n_0) + \log K$ in Section 8.
$u, u_j; v, v_j$	u is the negative-sign weight and u_j the relative corrections of (37); v is a marking variable in the proof of Theorem 4 and v_j the inverse coefficients of (40).
w, W	a signed permutation; $w = (e^t - 1)/q$ in the proof of Theorem 4; W is the positive Lambert function.
ℓ, x	ℓ indexes γ_ℓ and is $\log y$ in Section 8; x is the generating-function variable and a formal proxy for $1/n$.
$\kappa(q), \lambda$	the limiting fraction in the proof of Theorem 4; λ is the sparse-sign parameter (not the λ of [9]'s unbalanced core).

2 A natural weight for negative signs

For a signed permutation w , let $\text{neg}(w)$ count its negative entries and define

$$Z_n(u) = \sum_{\substack{w \in B_n \\ w \text{ has full support}}} u^{\text{neg}(w)}.$$

The usual sign-weighted extension of the classical generating function is

$$Z_n(u) = B_n(1+u), \quad n \geq 1. \quad (7)$$

Thus the parameter has a direct combinatorial meaning.

Here is an elementary verification. Decompose the underlying unsigned permutation into its indecomposable blocks. The signed permutation has full support precisely when its final unsigned block contains a negative entry. Indeed, a missing simple reflection corresponds to a proper unsigned block cut with every signed entry to its right positive; the sign-change reflection is missing when every entry is positive. This is also a direct consequence of the connectivity characterization in [1, Proposition 23].

If the final unsigned block has size k , there are $(n-k)!c_k$ underlying permutations. Writing $q = 1+u$, their allowed sign weight is $q^{n-k}(q^k - 1)$. Therefore

$$B_n(q) = \sum_{k=1}^n (n-k)!c_k q^{n-k}(q^k - 1) \quad (8)$$

$$= q^n n! - \sum_{k=1}^n c_k q^{n-k}(n-k)!. \quad (9)$$

In the second line we used $\sum_{k=1}^n c_k(n-k)! = n!$. In particular,

$$B_n(1) = 0 \quad (n \geq 1), \quad (10)$$

and $Z_n(u)$ has nonnegative integer coefficients.

3 The factorial expansion for connected permutations

Define the formal coefficients

$$\sum_{j \geq 0} \beta_j x^j = \frac{1}{g(x)^2} = (1 - C(x))^2. \quad (11)$$

The first values are 1, -2, -1, -4, -19, -110, -745.

Lemma 2. *For each fixed integer $R \geq 0$,*

$$\frac{c_n}{n!} = \sum_{j=0}^R \frac{\beta_j}{(n)_j} + O_R(n^{-R-1}). \quad (12)$$

Proof. The exact recurrence

$$c_n = n! - \sum_{k=1}^{n-1} c_k (n-k)!$$

first gives $0 < c_n \leq n!$ and $c_n/n! = 1 + O(n^{-1})$. Split its convolution into the two ends and the middle. For fixed R ,

$$\sum_{k=R+1}^{n-R-1} \binom{n}{k}^{-1} = O_R(n^{-R-1}). \quad (13)$$

For example, up to a sufficiently small fixed fraction of n , consecutive reciprocal-binomial terms decrease geometrically; in the central interval they are exponentially small. Since $c_k \leq k!$, this controls the middle part of the convolution.

An induction on R , using the already obtained lower-order expansions at the opposite endpoint, now determines the factorial-shift coefficients:

$$\beta_0 = 1, \quad \beta_r = -c_r - \sum_{j=1}^r j! \beta_{r-j} \quad (r \geq 1).$$

Their generating function is $(1 - C)/g = 1/g^2$, giving (11) and the stated remainder. $\square$

This factorial-composition phenomenon is classical; see Comtet [3] and Martin–Kearney [2]. The proof is included to make the error control self-contained. In ordinary inverse powers it gives

$$D(n) := \frac{c_n}{n!} \sim 1 - \frac{2}{n} - \frac{1}{n^2} - \frac{5}{n^3} - \frac{32}{n^4} - \frac{253}{n^5} - \dots. \quad (14)$$

4 Two exact endpoints and their uniform expansions

Set $h = \lfloor n/2 \rfloor$ and define the finite sums

$$\begin{aligned} L_n(q) &= 1 - \sum_{k=1}^h \frac{c_k}{q^k (n)_k}, \\ H_n(q) &= \sum_{j=0}^{n-h-1} \frac{q^j j! D(n-j)}{(n)_j}. \end{aligned} \quad (15)$$

Equation (9) becomes the exact identity

$$\frac{B_n(q)}{q^n n!} = L_n(q) - q^{-n} H_n(q). \quad (16)$$

Theorem 3. For every fixed R , each of $L_n(q)$ and $H_n(q)$ has an expansion through n^{-R} with error $O(n^{-R-1})$, uniformly on compact subsets of

$$\frac{1}{4} < |q| < 4.$$

Every fixed derivative in q has the corresponding uniform expansion. In particular the result holds on $[1, Q]$ for each fixed $Q < 4$.

Proof. The tail estimates reduce to sums of $Q^j / \binom{n}{j}$ for $j \leq n/2$ and $Q < 4$. Up to δn , choose $\delta > 0$ so that the consecutive ratio is at most $1/2$. Beyond this point, write the exponential rate as

$$t \log Q - \mathcal{H}(t), \quad \mathcal{H}(t) = -t \log t - (1-t) \log(1-t).$$

It is a convex function of t . At a sufficiently small δ and at $1/2$ its values are negative, since $\frac{1}{2} \log Q - \log 2 < 0$. The intervening tail is exponentially small. Removing the first $R+1$ terms consequently leaves $O_R(n^{-R-1})$. For L_n use $Q = \max |q|^{-1} < 4$; for H_n use $Q = \max |q| < 4$. Fixed derivatives introduce only fixed powers of the summation index and do not affect these conclusions. Finally apply Lemma 2 in the finitely many surviving terms of H_n . $\square$

With x a formal proxy for $1/n$, define

$$D(x) = \sum_{j \geq 0} \frac{\beta_j x^j}{\prod_{r=0}^{j-1} (1 - rx)}, \quad (17)$$

$$L(q, x) = 1 - \sum_{k \geq 1} \frac{c_k q^{-k} x^k}{\prod_{r=0}^{k-1} (1 - rx)}, \quad (18)$$

$$H(q, x) = \sum_{j \geq 0} \frac{q^j j! x^j}{\prod_{r=0}^{j-1} (1 - rx)} D\left(\frac{x}{1 - jx}\right). \quad (19)$$

Each fixed coefficient involves only finitely many terms, so these are effective all-order generators. Their first coefficients are

$$L(q, x) = 1 - \frac{x}{q} - \frac{x^2}{q^2} - \frac{q+3}{q^3} x^3 - \frac{q^2+9q+13}{q^4} x^4 + \dots, \quad (20)$$

$$\begin{aligned} H(q, x) = & 1 + (q-2)x + (2q^2 - 2q - 1)x^2 \\ & + (6q^3 - 2q^2 - 3q - 5)x^3 \\ & + (24q^4 + 6q^3 - 12q^2 - 9q - 32)x^4 + \dots. \end{aligned} \quad (21)$$

For an arbitrary fixed $q > 1$, including $q \geq 4$, the single main expansion (3) remains valid. Indeed the omitted original terms with $k > n/2$ in (9), normalized by $q^n n!$, total at most $nq^{-n/2}$. The first half is controlled by reciprocal-binomial tails. This observation does *not* extend the separate secondary-endpoint bounds to $q \geq 4$.

At $q = 1$, equations (10) and (16) imply $L_n(1) = H_n(1)$ exactly, hence $L(1, x) = H(1, x)$ as formal series. This cancellation is essential in the sparse-sign regime.

Exponential-scale qualification. The exact finite sums in (15) define the two pieces in (16). Replacing L_n by any fixed algebraic truncation leaves a polynomially small error, which is larger than q^{-n} . Thus the formula is not a claim that a fixed truncation of the main series alone resolves an exponentially small remainder. No optimal-truncation or complete infinite-transseries assertion is made.

5 The late coefficients as a positive Stirling transform

Using Stirling numbers of the second kind,

$$\frac{1}{(n)_k} = \sum_{m \geq k} \left\{ \begin{matrix} m-1 \\ k-1 \end{matrix} \right\} n^{-m}.$$

It follows from (18) that

$$p_m(q) = - \sum_{k=1}^m c_k q^{-k} \left\{ \begin{matrix} m-1 \\ k-1 \end{matrix} \right\}, \quad m \geq 1. \quad (22)$$

This also defines the rational-function extension of p_m to every $q \neq 0$. It is this extension that is used below for $0 < q \leq 1$; the original one-sector count asymptotic (3) is asserted only for $q > 1$. Since every summand in (22) is nonnegative and at least one is positive, $p_m(q) < 0$ for $q > 0$.

If $C_e(z) = \sum_{k \geq 1} c_k z^k / k!$, the exact shifted Borel transform in its disk of convergence is

$$\sum_{m \geq 1} \frac{p_m(q) t^{m-1}}{(m-1)!} = -\frac{1}{q} C'_e \left(\frac{e^t - 1}{q} \right). \quad (23)$$

The general shifted-Stirling and compound-Poisson construction is explicitly developed in [2]; it is prior methodology.

6 Every fixed late coefficient order

For $q > 0$, put

$$\tau = \log(1+q), \quad a = \frac{1+q}{q}, \quad A_2 = -\frac{q}{(1+q)^2}, \quad A_1 = \frac{q+2}{(q+1)^2}.$$

For $\ell \geq 0$, define

$$\gamma_\ell(q) = [\delta^\ell] \left\{ -\frac{1}{q} \sum_{j=2}^{\ell+2} \frac{(-1)^{j-1} \beta_j}{(j-2)!} [a(1-e^{-\delta})]^{j-2} \right\}. \quad (24)$$

Theorem 4. *For every fixed $J \geq 0$, as $m \rightarrow \infty$,*

$$\begin{aligned} p_m(q) &= A_2 m! \tau^{-m-1} + A_1 (m-1)! \tau^{-m} \\ &\quad + \sum_{\ell=0}^J \gamma_\ell(q) (-1)^{\ell+1} \ell! (m-\ell-2)! \tau^{\ell+1-m} \\ &\quad + O\left((m-J-3)! \tau^{J+2-m}\right). \end{aligned} \quad (25)$$

The remainder is uniform for q in any fixed compact subinterval of $(0, \infty)$.

Proof. We give a finite positive-transform proof, avoiding any assumption about analytic continuation of the full function C_e .

First substitute the factorial-shift expansion of Lemma 2 into (22). For any selected M , the error is bounded by a constant times $k!k^{-M}$. Consider the positive comparison weights

$$k!q^{-k} \left\{ \begin{matrix} m-1 \\ k-1 \end{matrix} \right\}.$$

When additionally marked by v^k , their exponential generating function is

$$\sum_{m \geq 1} \frac{t^{m-1}}{(m-1)!} \sum_{k=1}^m k! (v/q)^k \begin{Bmatrix} m-1 \\ k-1 \end{Bmatrix} = \frac{v/q}{(1 - (v/q)(e^t - 1))^2}. \quad (26)$$

For positive v its nearest singularity is the double pole

$$\tau(v) = \log(1 + q/v);$$

all other poles are $\tau(v) + 2\pi ir$, $r \in \mathbb{Z} \setminus \{0\}$. Residue extraction is uniform for v near 1 and q in a compact positive interval. Consequently the exponential scale is $\tau(v)^{-m}$. The usual exponential-moment bounds imply that k lies in $[\eta m, (1 - \eta)m]$ apart from an exponentially small fraction of the total weight, for some $\eta > 0$. Its limiting fraction is

$$\kappa(q) = \frac{q}{(q+1)\log(1+q)} \in (0, 1).$$

For clarity, the lower-tail bound uses $v < 1$ and the upper-tail bound uses $v > 1$; choosing the thresholds strictly below and above $\kappa(q)$ makes their exponential rates negative. Compactness gives a uniform choice. It follows that the transformed factorial-approximation error is $O(m^{-M})$ relative to the comparison sum. The finitely many small- k terms are negligible on the factorial scale.

Next evaluate each factorial-shift model exactly. For $j = 0$ and $j = 1$, the model generating functions are $(1 - w)^{-2}$ and $(1 - w)^{-1}$. For $j \geq 2$,

$$\sum_{k \geq j} \frac{(k-j)!}{(k-1)!} w^{k-1} = \frac{(-1)^{j-1}}{(j-2)!} (1-w)^{j-2} \log(1-w) + P_j(w), \quad (27)$$

where P_j is a polynomial. This identity follows by repeated integration, starting with $-\log(1-w)$ at $j = 2$.

Substitute $w = (e^t - 1)/q$ and $\delta = \tau - t$. Then

$$1 - w = a(1 - e^{-\delta}), \quad \log(1 - w) = \log \delta + \text{a function analytic at } \delta = 0.$$

The double and simple pole coefficients are A_2, A_1 , and the logarithmic coefficients are exactly (24). These finite model generating functions are explicit rational and logarithmic functions of e^t , whose nearest singularity is $t = \tau$. Coefficient transfer is applied only to these explicit functions. The analytic polynomial pieces become entire exponential polynomials and are negligible on the factorial scale.

Finally, for $m - 1 > \ell$,

$$(m-1)! [t^{m-1}] (\tau - t)^\ell \log(\tau - t) = (-1)^{\ell+1} \ell! (m - \ell - 2)! \tau^{\ell+1-m}. \quad (28)$$

The first omitted logarithmic term has the size of the remainder in (25). Taking $M \geq J + 4$ in the positive-transform estimate makes its error no larger. This proves the theorem. $\square$

The first explicit relative corrections are

$$\begin{aligned} \frac{p_m(q)}{A_2 m! \tau^{-m-1}} &= 1 - \frac{(q+2)\tau}{qm} - \frac{(q+1)^2 \tau^2}{q^2(m)_2} - \frac{4(q+1)^3 \tau^3}{q^3(m)_3} \\ &\quad - \frac{(q+1)^3 (23q+19) \tau^4}{q^4(m)_4} + O_q(m^{-5}). \end{aligned} \quad (29)$$

Putting $q = 2$ and multiplying by 2^m proves Theorem 1, including its normalization.

7 Sparse signs and a conditional Poisson law

Fix $\lambda > 0$ and give each negative sign weight $u = \lambda/n$. Write $x = 1/n$ formally.

Theorem 5. *For every fixed R , locally uniformly for complex λ ,*

$$\frac{B_n(1 + \lambda/n)}{n!} = \sum_{r=0}^R \frac{F_r(\lambda)}{n^r} + O_R(n^{-R-1}), \quad (30)$$

where the all-order coefficient generator is

$$\begin{aligned} \sum_{r \geq 0} F_r(\lambda) x^r &= e^\lambda \exp \left(\sum_{s \geq 2} \frac{(-1)^{s+1} \lambda^s}{s} x^{s-1} \right) L(1 + \lambda x, x) \\ &\quad - H(1 + \lambda x, x). \end{aligned} \quad (31)$$

On a compact disk containing zero, the remainder in (30) is in fact $O_R(|\lambda|n^{-R-1})$.

Proof. Apply the two separate endpoint expansions in Theorem 3 in a fixed complex neighborhood of $q = 1$, and expand $n \log(1 + \lambda/n)$. This proves (30) and (31), uniformly on compact complex sets. Every $F_r(0)$ is zero because $L(1, x) = H(1, x)$. The holomorphic remainder also vanishes at zero. Cauchy bounds on a slightly larger compact disk therefore improve its bound to $O_R(|\lambda|n^{-R-1})$. $\square$

The first coefficients are

$$F_0(\lambda) = e^\lambda - 1, \quad (32)$$

$$F_1(\lambda) = - \left(1 + \frac{\lambda^2}{2} \right) e^\lambda + 1, \quad (33)$$

$$F_2(\lambda) = e^\lambda \left(\frac{\lambda^4}{8} + \frac{\lambda^3}{3} + \frac{\lambda^2}{2} + \lambda - 1 \right) - \lambda + 1. \quad (34)$$

After division by $e^\lambda - 1$, the improved remainder gives a relative $O_R(n^{-R-1})$ bound uniformly for $0 < \lambda \leq \Lambda$.

Under the probability on full-support signed permutations proportional to $(\lambda/n)^{\text{neg}(w)}$, let K_n count negative signs. Its probability generating function is

$$\mathbb{E}[z^{K_n}] = \frac{B_n(1 + \lambda z/n)}{B_n(1 + \lambda/n)} \longrightarrow \frac{e^{\lambda z} - 1}{e^\lambda - 1}. \quad (35)$$

Thus K_n converges in distribution to a Poisson random variable of mean λ conditioned to be positive. In particular,

$$\mathbb{E}[K_n] \longrightarrow \frac{\lambda e^\lambda}{e^\lambda - 1}.$$

Dividing the two series from Theorem 5 gives every fixed correction order in (35), locally uniformly in z .

There is also an exact one-negative-sign check:

$$B'_n(1) = c_{n+1}. \quad (36)$$

Indeed the identity $x^2 g'(x) + (x-1)g(x) = -1$ implies

$$\frac{x g'(x)}{g(x)} = \frac{C(x)}{x} - 1.$$

The first one-negative counts are 1, 3, 13, 71, 461. Accordingly the derivatives $F'_r(0)$ reproduce the expansion of $c_{n+1}/(nn!)$.

8 Growth index inversion

The expansions above give three classes of positive sequences of the form

$$X_n = K\alpha^n \Gamma(n+1) \left(1 + \sum_{j=1}^R \frac{u_j}{n^j} + O(n^{-R-1}) \right). \quad (37)$$

The parameters are as follows:

- For $|a_n|$, $\alpha = 2/\log 3$ and $K = 2/(9\log 3)$, with u_j generated by (6).
- For $B_n(q)$ at fixed $q > 1$, $\alpha = q$, $K = 1$, and $u_j = p_j(q)$. For type D , $\alpha = 2$, $K = 1/2$, and the same algebraic coefficients $p_j(2)$ apply, with the exact correction (2) on the smaller factorial scale.
- For $B_n(1 + \lambda/n)$ at fixed $\lambda > 0$, $\alpha = 1$, $K = e^\lambda - 1$, and $u_j = F_j(\lambda)/(e^\lambda - 1)$.

The sequences are eventually increasing, since their consecutive ratios are asymptotic to $\alpha(n+1)$.

For $\ell = \log y$, let W be the positive Lambert function and put

$$n_0 = \frac{\ell}{W(\alpha\ell/e)}, \quad D_0 = \log(\alpha n_0), \quad A_0 = \frac{1}{2} \log(2\pi n_0) + \log K. \quad (38)$$

The first two inverse correction steps are

$$\begin{aligned} \nu(\ell) &= n_0 - \frac{A_0}{D_0} - \frac{A_0^2/(2D_0^2) - A_0/(2D_0) + v_1}{n_0 D_0} + O\left(\frac{(\log n_0)^2}{n_0^2}\right), \\ v_1 &= \frac{1}{12} + u_1. \end{aligned} \quad (39)$$

This follows by two Taylor–Newton steps applied to the logarithm of (37). Every subsequent fixed order is obtained by formal Newton iteration on

$$\ell = x(\log(\alpha x) - 1) + \frac{1}{2} \log(2\pi x) + \log K + \sum_{j \geq 1} v_j x^{-j}, \quad (40)$$

where v_j combines the Stirling log-Gamma coefficients with $\log(1 + \sum_{j \geq 1} u_j x^{-j})$.

Integer thresholds. If a sufficiently deep smooth truncation defines the real solution ν_R with logarithmic error $O(n^{-R-1})$, the true threshold

$$N(y) = \min\{n \geq n_* : X_n \geq y\}$$

obeys the enclosure

$$N(y) = \left\lceil \nu_R(\log y) + O\left(\frac{\nu_R^{-R-1}}{\log \nu_R}\right) \right\rceil. \quad (41)$$

Here n_* lies in the eventually increasing range. This notation describes a rounding ambiguity of the stated width. It does not assert an exact ceiling decision when y is too close to a sequence value, nor assume a canonical analytic interpolation of the discrete sequence.

[write] **Added 2 October 2026, batch 77: an instance of repository results.** The inversion of this section is an instance of the factorial-core inversion of ProveIt’s transseries volumes [8, 9], and no novelty is claimed for the method. The phase of (37) is $x(\log x + \log \alpha - 1) + O(\log x)$, so the seed (38) is the exact

solution of $X(\kappa \log X + d) = L$ with $\kappa = 1$, $d = \log \alpha - 1$ and $L = \ell$ in `p0:prop:factorial-core` of [8]: there $v = W_0((L/\kappa)e^{d/\kappa}) = W(\alpha\ell/e)$ and $X = e^{v-d/\kappa} = (e/\alpha)e^v = \ell/v = n_0$, and the core slope $\kappa \log X + d + \kappa = \log(\alpha n_0)$ is the D_0 above. It is equally the unbalanced core `t2:eq:unbalanced-core` of [9] ($\kappa = 1$, $\lambda = \log \alpha - 1$, with $L = \log y$ in place of $\log(y/C)$). In the variable $s = \log X + d$ that equation reads $s + \log s = \log \ell + d$, which is `p0:thm:lambert-core` with $a = b = 1$; since $b > 0$ its branch rule selects W_0 , the positive branch used here. The corrections (39) and the formal Newton iteration on (40) are the reversion about an admissible core, `p0:thm:core-reversion` of [8] (in [9], the scaled local reversion `t2:prop:scaled`); the linear-logarithmic specialization `p0:thm:lambert-centered` and the balanced inverse `t2:thm:balanced-inverse` do not apply directly, because the phase is $x \log x$ rather than $ax + b \log x$. The enclosure (41) is the separation condition, part (2) of `p0:thm:staircase`: the threshold is determined when the two ceilings at the ends of the stated width agree, and not otherwise. Part (1) of that theorem (exact rounding) needs an admissible interpolation of the sequence itself, which the smooth truncation ν_R is not; that is the qualification stated above. The generic staircase arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone function and give no formal status to anything in this section. Specific to this report are the three carriers of (37) with their coefficients u_j , which come from Theorem 1, from (3) with (22), and from Theorem 5.

9 Reproducibility and attribution

The accompanying offline Python scripts use exact integer recurrences and symbolic series. The verification includes the first displayed OEIS terms, weighted signed-permutation enumeration through $n = 6$, and large-order, endpoint, and sparse-sign numerical checks through 600. The coefficient generator produces five sparse-sign correction orders and six displayed late factorial-basis corrections. The numerical tests are consistency checks; the proofs and uniform remainders are given above.

The following boundaries are important. The B/D models and generating functions are established in [1]; their leading factorial asymptotics and several corrections are already recorded in [5, 6]. The connected-permutation expansion is classical, and [2] contains the general shifted-Stirling/compound-Poisson construction and the relevant integral representations. These are credited foundations. The specific late-growth statement in [4], its explicit all-fixed-order refinement, and the signed-weight crossover are the targets proved here. A bounded source search found no prior statement of this full specialized package; exhaustive historical priority is not asserted.

[write] Added 2 October 2026, batch 77. In the collection the two scripts and the delivered `build.sh` and `replay.sh` are in `code/`, and the three recorded outputs and `requirements.txt` in `data/`, all byte-identical to the delivery; the delivered PDF and the checksum ledger `SHA256SUMS` (verified at intake and retired) are not shipped and survive in the archive. `replay.sh` changes to its own directory and then calls `code/derive.py`, so in the collection layout (where it lives in `code/`) it does not find the scripts; `verify.py` writes `verification.json` into the `data/` directory next to its own parent, that is, over the shipped file; `build.sh` compiles the delivered file name `Coxeter_Full_Support_Asymptotics.tex`, which is shipped only as this `article.tex`. The report's `README.md` gives a rerun on a scratch copy that leaves the shipped files untouched.

References

- [1] N. Bergeron, C. Hohlweg, and M. Zabrocki, *Posets related to the connectivity set of Coxeter groups*, J. Algebra **303** (2006), 831–846. [arXiv:math/0509271v3](https://arxiv.org/abs/math/0509271v3). In particular Propositions 23–26.

- [2] R. J. Martin and M. J. Kearney, *Integral representation of certain combinatorial recurrences*, *Combinatorica* **35** (2015), 309–315. [doi:10.1007/s00493-014-3183-3](https://doi.org/10.1007/s00493-014-3183-3). The author manuscript gives the compound-Poisson construction and the type B/D integral representations.
- [3] L. Comtet, *Sur les coefficients de l'inverse de la série formelle $\sum n!t^n$* , *C. R. Acad. Sci. Paris A* **275** (1972), 569–572. Classical background; Lemma 2 is proved independently here.
- [4] V. Kotěšovec, [OEIS A260952](#), submitted 5 August 2015. The large-order conjecture and its $1/(2n)$ normalization. Accessed 2 October 2026.
- [5] M. Zabrocki and contributors, [OEIS A109253](#). Type B full-support counts and the posted asymptotic coefficients. Accessed 2 October 2026.
- [6] M. Zabrocki and contributors, [OEIS A112225](#). Type D full-support counts, with offset 2. Accessed 2 October 2026.
- [7] [write] ProveIt research-report collection, the A260700 report *All fixed order asymptotics for distinct parabolic double cosets* (batch 77, manuscript 17), `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a260700-parabolic-double-cosets`.
- [8] [write] ProveIt, *Transseries and inversion*, the transseries-and-inversion volume (`p0:thm:lambert-core`, `p0:prop:factorial-core`, `p0:thm:lambert-centered`, `p0:thm:core-reversion`, `p0:thm:staircase`), `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`.
- [9] [write] ProveIt, *Combinatorial transseries inverses* (`t2:eq:unbalanced-core`, `t2:prop:scaled`, `t2:thm:balanced-inverse`), `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`.